
Benjamin Otoo

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PROFESSIONAL SUMMARY

Clinical Scientist and microbiologist with experience spanning infectious disease research, molecular diagnostics, next-generation sequencing, and computational genomics. Experienced in *Mycobacterium tuberculosis* research, antimicrobial susceptibility testing, fungal pathogen research, clinical laboratory workflows, and NGS-based molecular testing. Develops bioinformatics and data-analysis workflows using Python, Linux, and genomic analysis tools, with additional experience building web-based applications and computational pipelines for biological data. Experienced in research coordination, laboratory training, mentorship, and communicating scientific findings across multidisciplinary environments.

TECHNICAL SKILLS

Pathogen & Molecular Biology: *Mycobacterium tuberculosis* culture, microbial culture and identification, antimicrobial susceptibility testing, molecular diagnostics, PCR, DNA/RNA extraction, NGS library preparation, clinical specimen processing, assay development, microscopy

Genomics & Bioinformatics: NGS data analysis, variant calling, variant annotation, VCF analysis, genomic data analysis, bioinformatics databases, sequencing QC, VEP, BWA, SAMtools, BCFtools, cyvcf2

Programming & Data Analysis: Python, SQL, Bash/Shell scripting, pandas, NumPy, Matplotlib, Plotly, reportlab, exploratory data analysis

Workflow & Computing: Linux/UNIX, Conda, Docker, Docker Compose, Git/GitHub, computational pipeline development

Software & Databases: PostgreSQL, Flask, Jinja, HTML/CSS, relational database design

Laboratory Quality & Validation: Clinical laboratory workflows, assay validation, analytical performance evaluation, QC/quality metrics, SOP development, CLIA/ISO-compliant laboratory practices

PROFESSIONAL EXPERIENCE

CLINICAL SCIENTIST / NGS SCIENTIST

Exact Sciences Laboratories — Madison, WI | May 2024–Present

- Perform next-generation sequencing workflows supporting molecular testing, cancer biomarker research, and assay development, including cfDNA, buffy coat, and FFPE DNA extraction and sequencing library preparation.
- Operate and maintain Illumina NovaSeq 6000 and NovaSeq X sequencing platforms, troubleshooting workflow and instrument issues to maintain data quality and throughput.

- Perform DNA quantification and quality assessment using Qubit and TapeStation, interpreting QC metrics to determine sample and library suitability for sequencing.
- Analyze experimental and sequencing datasets to evaluate assay performance, identify data-quality issues, and support research and development of molecular diagnostic assays.
- Collaborate across laboratory and research teams to troubleshoot experimental workflows, interpret results, and communicate findings.

PATHOGEN & INFECTIOUS DISEASE RESEARCH

LABORATORY MANAGER

University of Wisconsin–Madison, Keller Lab — Madison, WI | Sep 2022–Dec 2023

- Managed day-to-day laboratory operations supporting research on fungal pathogen biology and secondary metabolism.
- Conducted research investigating ZfpA-dependent quorum sensing, morphological development, and secondary metabolism in *Aspergillus flavus*.
- Developed and maintained standard operating procedures for molecular and microbiological assays, supporting consistent and reproducible laboratory workflows.
- Coordinated experimental workflows, laboratory resources, and research activities across multiple projects.
- Mentored undergraduate and graduate researchers in laboratory techniques, experimental workflows, and research practices.

RESEARCH ASSISTANT

Korle Bu Teaching Hospital — Accra, Ghana | Dec 2021–Aug 2022

- Conducted clinical research investigating fungal infections among patients with tuberculosis, HIV, and hematological disorders.
- Supported research, development and implementation of molecular diagnostic protocols for detection of invasive fungal diseases.
- Participated in patient recruitment, clinical data and specimen collection, laboratory testing, and research data analysis.
- Contributed to study documentation, organization of laboratory results, and interpretation of findings in a clinical infectious disease research setting.

RESEARCH ASSISTANT

Noguchi Memorial Institute for Medical Research — Accra, Ghana | Sep 2020–Nov 2021

- Investigated the relationship between diabetes mellitus and tuberculosis treatment, with a focus on antimicrobial resistance associated with prolonged TB treatment and the gut microbiome.
- Performed *Mycobacterium tuberculosis* culture, microbiological culture, and antimicrobial susceptibility testing to support infectious disease research.
- Participated in patient recruitment and longitudinal monitoring and collected clinical and research data and specimens.
- Performed urinalysis and maintained laboratory records and study data in support of clinical research activities.
- Contributed to organization and reporting of research findings from tuberculosis and antimicrobial resistance studies.

BIOINFORMATICS & SOFTWARE PROJECTS

ONCOTRACE — VCF ANALYSIS & GENOMIC DATA PLATFORM

Python | Flask | PostgreSQL | Docker | Linux | BCFtools | VEP | cyvcf2 | Plotly

- Developed a researcher-facing web application for uploading, processing, analyzing, and visualizing VCF-based genomic data.
- Built Python workflows for VCF parsing, variant-level analysis, sequencing quality metrics, and variant annotation.
- Integrated BCFtools and Ensembl VEP to support variant statistics, filtering, and functional annotation of genomic variants.
- Developed PostgreSQL data models linking projects, patients, biological samples, VCF files, variants, observations, and annotations for traceable genomic data management.
- Implemented quantitative analyses including depth of coverage, variant allele frequency, mapping quality, and variant quality metrics.
- Developed interactive Plotly visualizations for chromosome-level variant distribution, variant classes, and sequencing quality metrics.
- Containerized the application and supporting database and bioinformatics services using Docker and Docker Compose.

QCFAST — FASTQ QUALITY-CONTROL APPLICATION

Python | Flask | Docker | pandas | NumPy | Matplotlib | ReportLab

- Developed a FASTQ quality-control application for evaluating sequencing data and generating quality metrics and reports.
- Implemented analysis of read count, total bases, read length, GC content, N content, mean quality, Q20, and Q30 metrics.
- Supported paired-end FASTQ analysis with read-pair validation and FASTQ/gzip input handling.
- Developed sequencing-quality visualizations and automated PDF reporting through a Flask-based web interface.
- Containerized the application with Docker and provided command-line functionality for reproducible QC analysis.

EDUCATION

UNIVERSITY OF WISCONSIN—MADISON — Madison, WI

Master of Science, Microbiology | May 2024

GPA: 3.73/4.00

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY — Kumasi, Ghana

Bachelor of Science, Biological Sciences | May 2020

First-Class Honors | GPA: 3.73/4.00

SELECTED PUBLICATIONS

Otoo, B., et al. "ZfpA-Dependent Quorum Sensing Shifts in Morphology and Secondary Metabolism in *Aspergillus flavus*." *Environmental Microbiology*, 2025.

Ocansey, B.K., Otoo, B., et al. "Chronic pulmonary aspergillosis is common among patients with presumed tuberculosis relapse in Ghana." *Medical Mycology*, 2022.

Ocansey, B.K., Otoo, B., et al. "Cryptococcal and *Histoplasma* Antigen Screening Among People With Human Immunodeficiency Virus in Ghana and Comparative Analysis of OIdx *Histoplasma* Lateral Flow..." *Open Forum Infectious Diseases*, 2022.

Ocansey, B.K., **Otoo, B.**, et al. "Importance of *Aspergillus*-Specific Antibody Screening for Diagnosis of Chronic Pulmonary Aspergillosis after Tuberculosis Treatment: A Prospective Follow-Up..." *Journal of Fungi*, 2023.

Link to google scholar: <https://scholar.google.com/citations?user=Vs-npXoAAAAJ&hl=en>